An elementary VSL lane-drop task: model and sensitivity

1. Research question

Can the distribution of traffic between continuing lanes change useful common variable-speed-limit (VSL) advice, even when ending-lane traffic and total continuing traffic agree? The task connects lane information to receiving, merge service and complete travel time, then differentiates that same causal chain for control.

2. Task and global assumptions

A permanent three-to-two drop has ending lane E, adjacent continuing lane A and far continuing lane B. Preparation ends at L_p; a terminal of length b ends at the choke $L = L_p + b$. At time zero, independent lane densities $(e_0, a_0, b_0) \geq 0$ are uniform on the common footprint $[0, \ell]$, $0 < \ell \leq L_p$. Set $r = e_0 + a_0$ and total mass $M = \ell(e_0 + a_0 + b_0) > 0$. The terminal and recovery history start empty; there are no later external arrivals. Choose one constant common advice $v \in [v_-, v_+]$, $0 < v_- < v_+$, with fixed initial state, geometry and physical parameters. The objective counts every initial unit's time until it crosses the choke.

Global assumptions. Traffic is continuous mass; the remaining preparation body is homogeneous. Unblocked motion follows the common advice, with fixed backward-wave parameter w and jam density k_j. Only E-to-A transfer occurs, before the terminal and at the choke; B has separate service. Point service uses current-state priority/cooperation and representative free/queued speed marks. Forced E entries alone create recovery loss, retained as age/mark history behind one shared service budget.

Triangular-diagram and cumulative-boundary ingredients follow LTM [1]; the advice-response family is selected here. Define

$$K = wk_j, \quad k_c(v) = \frac{K}{w+v}, \quad C(v) = vk_c(v), \qquad c(u) = \frac{Ku}{w+u}, \quad U(g) = \frac{wg}{K-g}. \tag{1}$$

R and R_B are undisturbed A/E and B service ceilings, respectively. Use positive physical scales and the sufficient operating domain

$$e_0, a_0, b_0 \leq k_c(v_+), \qquad v_+ r < R \leq C(v_-), \qquad v_+ b_0 \leq R_B \leq C(v_-), \qquad M < k_j b. \tag{2}$$

It keeps preparation free, terminal receiving available and B unqueued. Work and recovery choices are specified below; $[x]_+ = \max(x, 0)$.

3. Traffic model

Preparation determines lane-specific entries; terminal travel determines eligibility; forced service and recovery then update each other causally.

3.1. Pre-choke merging. Let $e(t), a(t)$ be the remaining preparation body's E/A densities. Own A flow uses receiving first. E transfer is limited by its donor offer and by remaining adjacent service:

$$S(v,a) = \min\{C(v), w(k_j - a)\}, \qquad f = \min\left\{ve, \frac{S(v,a) - va}{\gamma}\right\}, \qquad \dot{e} = -f/h, \quad \dot{a} = f/h. \tag{3}$$

Here $h > 0$ is a physical interaction length, $\gamma \geq 1$ is a receiving-work factor, and f/h is the lateral density source. Thus $e + a = r$ and b_0 stays constant. Donor-limited decay is exponential per traveled distance; advice affects preparation through normalized receiving headroom $k_c(v)$. *Adapted:* receiving work and own-lane priority draw on Carey et al. [2]; homogeneous exchange and its length scale are defined here.

3.2. Finite terminal and the merging point. For $j \in \{E, A, B\}$, let F_j, P_j, Y_j be cumulative terminal entry, service-eligible mass and departure. During $(L_p - \ell)/v < t < L_p/v$,

$$\dot{F}_E = ve(t), \quad \dot{F}_A = v[r - e(t)], \quad \dot{F}_B = vb_0; \qquad P_j(t) = F_j(t - b/v). \tag{4}$$

Entry rates vanish outside that interval; negative-time counts are zero. Define physical terminal stock $N_j = F_j - Y_j$, eligible waiting mass $q_j = P_j - Y_j$, offer $\lambda_j = \dot{P}_j$ and discharge $g_j = \dot{Y}_j$. Departures satisfy $0 \leq Y_j \leq P_j \leq F_j$, with $\dot{q}_j = \lambda_j - g_j$. Domain (2) protects finite storage and gives $Y_B = P_B$. Early-merged E becomes ordinary A; residual E travels through the terminal and merges at the choke. *Reused/adapted:* boundary travel and storage follow LTM [1]; the preparation cutoff and terminal handoff are defined here.

3.3. Forced merging and service sharing. Use free marks $u_j = v$ and queued marks $u_j = U(g_j)$ for $j = A, E$. A costs one work unit per mass; E's maneuver cost increases with adjacent flow and a positive speed difference:

$$c_E = c_0 + \chi g_A[u_A - u_E]_+, \qquad g_A + c_E g_E \le \mathcal{B}, \qquad H(g; g_A, u_A) = g\{c_0 + \chi g_A[u_A - U(g)]_+\}. \tag{5}$$

$c_0 \ge 1$ is baseline E work and $\beta \ge 0$ sets mismatch strength. Set $\chi = \beta/[C(v_+)v_+]$ once and hold it fixed. Require $c_0 > \beta(1 + v_+/w)$ so each feasible root $H(g) = x$ is unique. Here x is work allocated to E. A fitting free A offer has priority; queued A cooperates through equal work and unused-work redistribution:

A, E status	Selected service and free-admission condition
Free, free	$g_A = \lambda_A,\ g_E = \lambda_E$ if $\lambda_A + c_0\lambda_E \le \mathcal{B}$.
Free, queued	$g_A = \lambda_A \le \mathcal{B}, \quad H(g_E; \lambda_A, v) = \mathcal{B} - \lambda_A$.
Queued, queued	$g_A = \mathcal{B}/2, \quad H(g_E; \mathcal{B}/2, U(\mathcal{B}/2)) = \mathcal{B}/2$.
Queued, free	$g_E = \lambda_E,\ g_A = \mathcal{B} - c_0\lambda_E$ if $c_0\lambda_E \le \mathcal{B}/2$.

A strict free-service shortfall creates a queue. At an actual zero hit, retest free admission; equality retains feasible free continuation. Use right-hand arrivals at jumps and retain recovery history across transitions. *Defined here:* the mismatch formula, representative marks and sharing rule complete the receiving-work interpretation of [2].

3.4. Recovery and the shared budget. A forced E entry of mark u leaves an age-dependent loss. For age $\theta \ge 0$ and recovery acceleration $a_r > 0$, define

$$V(\theta; u) = \sqrt{(w + u)^2 + 2wa_r\theta} - w, \qquad D(\theta; u) = [R - c(V(\theta; u))]_+. \tag{6}$$

V is an equivalent boundary-speed quantity; D is remaining lost service. Let $T(\xi)$ be the admission time of E mass rank ξ and fix $m = 1$ vehicle-equivalent. The recovery reading and available budget are

$$p_{\mathrm{rec}}(t) = \frac{1}{m}\int_{[Y_E(t)-m]_+}^{Y_E(t)} D\big(t - T(\xi); u_E(T(\xi))\big)\,\mathrm{d}\xi, \qquad \mathcal{B}(t) = [R - p_{\mathrm{rec}}(t)]_+. \tag{7}$$

All retained profiles age; after the window fills, new E replaces oldest mass weight. The fixed denominator also applies before fill. A consumes the shared budget while E supplies the profiles; their predicted age/mark history updates the account. *Adapted:* the isolated profile draws on Leclercq et al. [3]. *Defined here:* capping, the E-only source and recent-mass renewal.

4. How the task tests information

Compare the same initial E mass and continuing total with different A/B splits. A busier A can restrict preparation, leaving more E for forced work and recovery. Advice trades these effects against ordinary travel time. With a fixed horizon H_* beyond cohort clearance, minimize complete TST:

$$J(v) = \int_0^{H_*}\left[M - \sum_j Y_j(t)\right]\mathrm{d}t = \frac{M(L - \ell/2)}{v} + \int_0^{H_*}(q_A + q_E)\,\mathrm{d}t. \tag{8}$$

The predictors share geometry and physical parameters and may use all inference from admitted predecision histories: aggregate density, flow, speed and cumulative counts, including any permitted E-resolved history. One knows initial E and combined A+B mass; the other knows each lane. An equal-information compact peer adds just initial A mass to the former, reconstructing all three initial densities and the same dynamics.

5. Complete sensitivity

The model-specific sensitivities are derived here at fixed initial traffic and physical parameters. Carry preparation, moving entry/eligibility clocks, service/marks, recovery aging/retirement and queue-event times through the full objective. A subscript v denotes advice sensitivity. On smooth branches,

$$g_{E,v} = \frac{x_v - H_{g_A} g_{A,v} - H_{u_A} u_{A,v}}{H_g}, \qquad H_g > 0. \tag{9}$$

For a filled window with $g_E(s_m) > 0$, let $Y_E(s_m) = Y_E(t) - m$ and $Z_E = Y_{E,v}$. The cutoff relation below is derived here. Differentiate (7) including marks, flows, retirement and source events. For an isolated transverse guard G along the history-aware trajectory, the remaining event corrections for continuous traffic coordinates X follow [4]:

$$\begin{gathered} g_E(s_m) s_{m,v} = Z_E(t) - Z_E(s_m), \qquad \tau_v = -G_v^- / G_t^-, \\ X_v^+ = X_v^- + (\dot{X}^- - \dot{X}^+)\tau_v. \end{gathered} \tag{10}$$

Thus the complete derivative, on a fixed regular event/history sequence, is

$$J'(v) = -\int_0^{H_*} \sum_j Y_{j,v}(t)\,\mathrm{d}t = -\frac{M(L - \ell/2)}{v^2} + \int_0^{H_*} (q_{A,v} + q_{E,v})\,\mathrm{d}t. \tag{11}$$

References

[1] W.-L. Jin (2015). Continuous formulations and analytical properties of the link transmission model.

[2] M. Carey, C. Balijepalli and D. Watling (2013 manuscript). Extending the Cell Transmission Model to Multiple Lanes and Lane-Changing.

[3] L. Leclercq, V. L. Knoop, F. Marczak and S. P. Hoogendoorn (2016). Capacity drops at merges: New analytical investigations.

[4] N. J. Kong, J. J. Payne, J. Zhu and A. M. Johnson. Saltation Matrices: The Essential Tool for Linearizing Hybrid Dynamical Systems, arXiv:2306.06862.